

6(a)(i)	$P = VI$	C1
	$I = 36 / 8.0$ $= 4.5 \text{ A}$	A1
6(a)(ii)	charge $= 4.5 \times 50$ $= 225$	C1
	number $= 225 / 1.6 \times 10^{-19}$ $= 1.4 \times 10^{21}$	A1
6(a)(iii)	$R = V^2 / P$ or $R = V / I$ or $R = P / I^2$ $= 8.0^2 / 36$ or $= 8.0 / 4.5$ or $= 36 / 4.5^2$	C1
	$= 1.8 \Omega$	A1

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Question	Answer	Marks
6(b)	$R = \rho L / A$	C1
	$L = (1.8 \times 0.25 \times 10^{-6}) / 1.4 \times 10^{-6}$ = 0.32 m	A1
6(c)	(larger cross-sectional area, same length, same resistivity and so) less resistance	M1
	(same p.d. and more current so) more power (dissipated)	A1
6(d)	current (in wire) is the same	M1
	(same p.d. across wire so) power stays the same	A1

Question	Answer	Marks
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